



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE-3.0

11-07-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Advanced)

TEST DATE : 11-07-2021

ANSWER KEY**PAPER-1****PHYSICS**

SEC-1	Q.	1	2	3	4	5	
	A.	A	D	A	B	C	
SEC-2	Q.	1	2	3	4	5	6
	A.	A,D	A,D	A,D	A,B	A,B,D	A,C
SEC-3	Q.	1	2	3	4		
	A.	B	B	A	B		
SEC-4	Q.	1	2	3	4		
	A.	312	232	312	123		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	
	A.	B	C	A	D	D	
SEC-2	Q.	1	2	3	4	5	6
	A.	A,D	A,C,D	B,C	A,B,C,D	A,C,D	A,B,C
SEC-3	Q.	1	2	3	4		
	A.	C	C	D	B		
SEC-4	Q.	1	2	3	4		
	A.	231	321	213	312		

MATHEMATICS

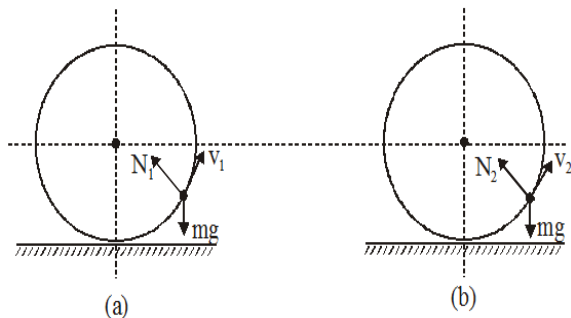
SEC-1	Q.	1	2	3	4	5	
	A.	A	A	D	B	B	
SEC-2	Q.	1	2	3	4	5	6
	A.	A	A,B,C,D	B,C,D	A,B,C	A,B	A,B,D
SEC-3	Q.	1	2	3	4		
	A.	B	D	A	B		
SEC-4	Q.	1	2	3	4		
	A.	231	312	132	231		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (A)



F.B.D in two cases is as shown in case (b) normal will be less and in case (b) work done by magnetic field is zero so height in two cases will be same.

2. Ans (D)

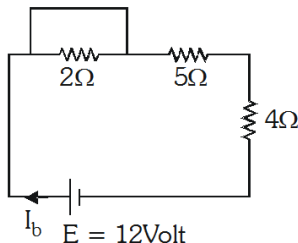
Torque due to magnetic force is zero and the resultant magnetic force = 0.

Thus there is no effect of magnetic field, on the motion of the block.

3. Ans (A)

For $t = 0$

$$I_b = \frac{12}{9} = \frac{4}{3} \text{ A}$$



4. Ans (B)

Correct reason is that the spectrum is shifted as temperature is increased. (Wein's displacement law)

PHYSICS

SEC-2

1. Ans (A,D)

$$\Delta \ell = \frac{F \ell_0}{A y}$$

$$\Rightarrow \frac{\ell_0}{A} = \frac{\ell'_0}{A'}$$

$$\frac{0.6}{15} = \frac{0.4}{A'} \Rightarrow A' = 10 \text{ cm}^2$$

$$\text{stress} = \frac{F}{A} \Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{A_2}{A_1} = \frac{10}{15}$$

$$\frac{\Delta r}{r} = -\sigma \frac{\Delta \ell}{\ell}$$

$$\Delta r = -\sigma \frac{\Delta \ell}{\ell} \times r = -\sigma \Delta \ell \frac{\sqrt{A}}{\ell}$$

$$\Delta r_{\text{left}} = \frac{\sqrt{15}}{0.6} = \sqrt{\frac{1500}{0.36}}$$

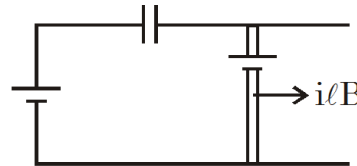
$$\Delta r_{\text{right}} = \frac{\sqrt{10}}{0.4} = \sqrt{\frac{1000}{0.16}}$$

2. Ans (A,D)

$$d = \frac{1}{2} a t^2 \Rightarrow d = \frac{1}{2} \frac{qV}{md} t^2$$

$$T = \frac{2\pi m}{qB} \Rightarrow N = \frac{t}{T}$$

3. Ans (A,D)



$$\varepsilon - B\ell v - \frac{q}{c} - iR = 0$$

$$i\ell B = \frac{mdv}{dt}$$

$$\ell Bq = mv$$

$$\varepsilon - B\ell v - \frac{mv}{B\ell c} = \frac{mR}{B\ell} \frac{dv}{dt}$$

$$a = 0$$

$$\Rightarrow v = \frac{\varepsilon}{B\ell + \frac{m}{B\ell c}} = \frac{\varepsilon B\ell c}{B^2 \ell^2 c + m}$$

$$q = \frac{mv}{B\ell}$$

4. Ans (A,B)

$$i = \frac{v_0}{\sqrt{(x_L - x_C)^2 + 100}}$$

$$V = i \times \sqrt{100 + x_C^2}$$

$$50\sqrt{5} = 5\sqrt{100 + x_C^2}$$

$$x_C = 20 = \frac{10^6}{\omega \times 50}$$

$$\omega = 10^3 \text{ rad/s}$$

$$\omega L = 20 \times 10^{-3} \times 10^3 = 20$$

$\Rightarrow$ Resonance

$$P = i^2 R = 250 \text{ W}$$

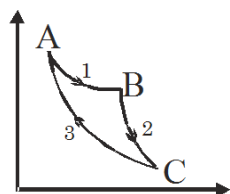
$$V = iR = 50 \text{ V}$$

5. Ans (A,B,D)

for Cycle

$$\Delta U_1 = 0$$

$$\Delta U_2 = -\Delta W_2 = -250 \text{ J}$$



$$\Delta U_3 = 250 \text{ J}$$

$$\Delta Q_3 = -50 = \Delta U_3 + \Delta W_3$$

$$\Delta Q_3 = -50 = 250 + \Delta W_3$$

$$\Delta W_3 = -300 \text{ J}$$

$$\Delta W = 600 + 250 - 300 = 550 \text{ J}$$

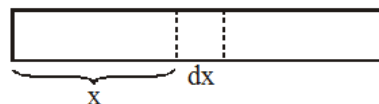
6. Ans (A,C)

stress is $\frac{F}{A}$, $\therefore$ it will be same at all points.

Young's modulus on element dx will be

$$Y_0 + \frac{Y_0}{L}x$$

let extension in dx be dy



$$\text{then } \frac{F}{A} = \left(Y_0 + \frac{Y_0 x}{L} \right) \frac{dy}{dx}$$

$$\int_0^L \frac{FL}{A} \frac{dx}{(Y_0 L + Y_0 x)} = \int_0^{\Delta L} dy$$

ΔL : total elongation in rod.

PHYSICS

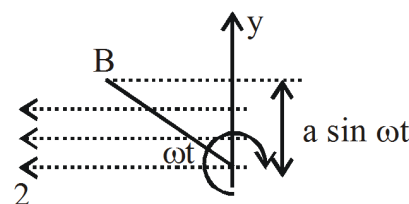
SEC-3

3. Ans (A)

From the diagram flux through the given loop

is same as the flux through a rectangular loop

having width $a \sin \omega t$ in xy frame.



$$\phi = \int_0^{a \sin \omega t} \frac{B_0}{a} y (a dy) = \frac{B_0}{2} a^2 \sin^2 \omega t$$

$$E = -\frac{B_0 a^2 \omega}{2} \sin 2\omega t$$

4. Ans (B)

Input power = Out put power (ω is constant)

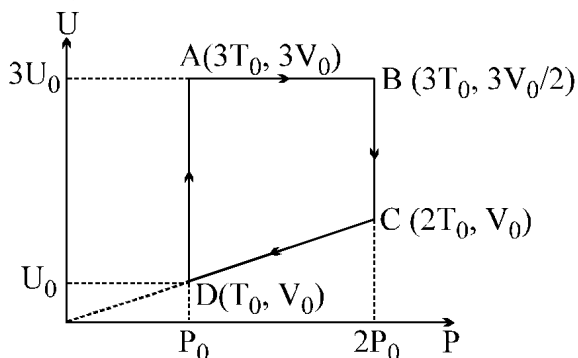
$$\tau \omega = \frac{\epsilon^2}{R}$$

$$\tau = \frac{B_0^2 a^4 \omega}{4R} \sin^2 2\omega t$$

PHYSICS

SEC-4

3. Ans (312)



As U (Internal Energy) is proportional to 'T'

If $V_D = V_0$ then $V_A = 3V_0$, $V_B = 3V_0/2$, $V_C = V_0$

Also if $T_D = T_0$, then $T_A = 3T_0$, $T_B = 3T_0$,

$T_C = 2T_0$

A → B isothermal $\Rightarrow \frac{V_B}{V_A} = \frac{1}{2}$

$$\therefore W_{AB} = nRT \ln\left(\frac{1}{2}\right) = -\frac{2}{3} \left(\frac{3}{2}RT\right) \ln 2$$

$$= -\frac{2}{3} \cdot 3U_0 (0.3010) = -0.602 U_0$$

B → C isobaric

$$W_{BC} = 2P_0(V_C - V_B) = -P_0V_0 = -RT_0 = -\frac{2}{3} U_0$$

$$= -0.67U_0$$

C → D: isochoric $\Rightarrow W_{CD} = 0$

D → A: isobaric

$$W_{DA} = P_0(V_A - V_D) = 2P_0V_0 = \frac{4}{3} \left(\frac{3}{2}RT_0\right)$$

$$= \frac{4}{3} U_0 = 1.33U_0$$

4. Ans (123)

When A is connected then heat flow from A to B to C to environment, so

$$T_A > T_B > T_C > T_0$$

Because A get continuous supply of heat, so temperature of A rises up which can not get value infinity so heat flow from A to environment.

when B is connected

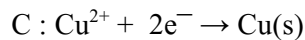
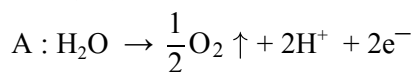
$$T_A = T_B > T_C > T_0$$

direction of heat flow should be from B to environment.

CHEMISTRY

SEC-1

1. Ans (B)



$$q = i \times t = 1.5 \times 10 \times 60 = 900 \text{ C}$$

193000C will produce 80 gm loss

$$\therefore 900 \text{ C will produce } \frac{80}{193000} \times 900 \text{ gm loss}$$

$$= 0.373 \text{ gm loss}$$

2. Ans (C)

$$C = \frac{\sqrt{C}}{10^3} \text{ K}$$

$$\Lambda_m = \frac{\kappa}{10^3 C} (\text{S-m}^2\text{-mol}^{-1})$$

$$10^{-3}M \quad 0.0316 \quad 1.237 \times 10^{-2}$$

$$12.37 \times 10^{-3}$$

$$10^{-2}M \quad 0.1000 \quad 11.85 \times 10^{-2}$$

$$11.85 \times 10^{-3}$$

$$\frac{\Lambda_m^\infty - 12.37 \times 10^{-3}}{0 - 0.0316} = \frac{12.37 \times 10^{-3} - 11.85 \times 10^{-3}}{0.0316 - 0.100}$$

$$\Lambda_m^\infty = 12.61 \times 10^{-3} \text{ S-m}^2\text{-mol}^{-1}$$

$$= 126.1 \text{ S-cm}^2\text{-mol}^{-1}$$

MATHEMATICS

SEC-1

1. Ans (A)

$$\{2, 2^2, 2^3, \dots, 2^{25}\}$$

$$\text{Total} = 25 \times 24$$

$$a = 2^{\lambda_1} \text{ \& } b = 2^{\lambda_2}$$

$$\lambda_1, \lambda_2 \in \{1, 2, \dots, 25\}$$

$$\log_a b = \log_2 \lambda_1 2^{\lambda_2} = \frac{\lambda_2}{\lambda_1} \in I$$

$$\text{fav} = \frac{24}{(\lambda_1=1)} + \frac{11}{(\lambda_1=2)} + \frac{7}{(\lambda_1=3)} + \frac{5}{(\lambda_1=4)}$$

$$\frac{4}{(\lambda_1=5)} + \frac{3}{(\lambda_1=6)} + \frac{2}{(\lambda_1=7)} + \frac{2}{(\lambda_1=8)}$$

$$\frac{1}{(\lambda_1=9)} + \frac{1}{(\lambda_1=10)} + \frac{1}{(\lambda_1=11)} + \frac{1}{(\lambda_1=12)}$$

$$62 = \text{fav}$$

$$P = \frac{\text{fav}}{\text{total}} = \frac{62}{600} = \frac{31}{300}$$

2. **Ans (A)**

$$\sum_{k=r}^n {}^k C_r = {}^r C_r + {}^{r+1} C_r + \dots + {}^n C_r$$

$$= {}^{r+1} C_{r+1} + {}^{r+1} C_r + {}^{r+2} C_r + \dots + {}^n C_r$$

$$= {}^{r+2} C_{r+1} + {}^{r+2} C_r + \dots + {}^n C_r$$

$$\Rightarrow {}^n C_{r+1} + {}^n C_r = {}^{n+1} C_{r+1}$$

$$f(n) = \sum_{r=0}^n {}^{n+1} C_{r+1} = {}^{n+1} C_1 + {}^{n+1} C_2 + \dots + {}^{n+1} C_{n+1} = 2^{n+1} - 1$$

$$f(9) = 2^{10} - 1 = 1023$$

3. **Ans (D)**

Let $Y_0 = \{7, 14, \dots, 98\} \Rightarrow 14$ element (7λ)
 $Y_1 = \{1, 8, \dots, 99\} \Rightarrow 15$ element ($7\lambda + 1$)
 $Y_2 = \{2, 9, \dots, 100\} \Rightarrow 15$ element ($7\lambda + 2$)
 $Y_3 = \{3, 10, \dots, 94\} \Rightarrow 14$ element ($7\lambda + 3$)
 $Y_4 = \{4, 11, 18, \dots, 95\} \Rightarrow 14$ element ($7\lambda + 4$)
 $Y_5 = \{5, 12, \dots, 96\} \Rightarrow 14$ element ($7\lambda + 5$)
 $Y_6 = \{6, 13, \dots, 97\} \Rightarrow 14$ element ($7\lambda + 1$)
 largest Y will consist of element of Y_0 (only 1 element), Y_1, Y_2 and one of Y_3 and Y_4
 $\Rightarrow 1 + 15 + 15 + 14 = 45$

4. **Ans (B)**

$(1 + x + x^2 + x^3 + x^4)^{100}$, Number of distinct terms = 401
 $(a_1 + a_2 + \dots + a_m)^n$:
 $r_1 + r_2 + \dots + r_m = n$, number of terms
 $= {}^{m+n-1} C_n$

5. **Ans (B)**

$$\frac{3}{4} \leq P(A \cup B) \leq 1 \text{ and } \frac{1}{8} \leq P(A \cap B) \leq \frac{3}{8}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

$$\frac{7}{8} \leq P(A) + P(B) \leq \frac{11}{8}$$

MATHEMATICS

SEC-2

1. **Ans (A)**

$(mn)!$
 (mn) things divided into m-equal groups each containing n-thing = No. of ways

$$\frac{(mn)!}{(n!)^m \cdot m!} \in I$$

(mn) thing divided into n-equal group each containing m-things = No. of ways

$$\frac{(mn)!}{(m!)^n \cdot n!} \in I$$

$$\frac{((mn)!)^2}{(n!)^m (m!)^n} \in I$$

$$\frac{((mn)!)^2}{(m!)^{n+1} (n!)^{m+1}} \in I$$

$$m = 1, n = 3, mn! = 3! = 6$$

$$(mn!)^2 = 36$$

$$(B) 6^4$$

$$(C) 6^3$$

$$(D) 6^4$$

2. **Ans (A,B,C,D)**

$$(x^3 + x^5)(2x^6 + 3x^4 - 1)^{20} = a_0 + a_1x + a_2x^2 + \dots + a_{125}x^{125}$$

In the expansion of $(x^3 + x^5)(2x^6 + 3x^4 - 1)^{20}$ all even powers of x is obviously zero.

Put $x = 1$, and $x = -1$ and subtract we get

$$2(a_1 + a_3 + \dots + a_{125}) = 2.4^{20} - (-2).4^{20}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{125} = 4.4^{20} = 2^{41}$$

$$\text{and } a_{125} = 2^{20} \Rightarrow a_1 + a_3 + \dots + a_{123} = 2^{41} - 2^{20}$$

on comparing the coefficient

$$a_0 = 0, a_1 = 0, a_2 = 0, a_3 = 1$$

3. **Ans (B,C,D)**

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$$

$$\Rightarrow (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = |z_1|^2 + |z_2|^2$$

$$\Rightarrow z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2 = |z_1|^2 + |z_2|^2$$

$$\Rightarrow z_1\bar{z}_2 + z_2\bar{z}_1 = 0 \Rightarrow \frac{z_1}{z_2} + \overline{\left(\frac{z_1}{z_2}\right)} = 0$$

$$\Rightarrow \frac{z_1}{z_2} \text{ is purely imaginary}$$

$$\Rightarrow \arg\left(\frac{z_1}{z_2}\right) = \pm \frac{\pi}{2}$$

4. Ans (A,B,C)

$$(x^n - 1) = (x - 1)(x - w)(x - w^2)(x - w^3) \dots (x - w^{n-1}) \dots (i)$$

$$\Rightarrow (1 + x + \dots + x^{n-1})$$

$$= (x - w)(x - w^2) \dots (x - w^{n-1}) \dots (ii)$$

$$\text{Put } x = 1 \text{ in (ii)} (1 - w)(1 - w^2) \dots (1 - w^{n-1}) = n$$

$$\text{Put } x = 4 \text{ in (i)}$$

$$\frac{4^n - 1}{3} = (4 - w)(4 - w^2) \dots (4 - w^{n-1})$$

$$\log(1 + x + \dots + x^{n-1})$$

$$= \log(x - w) + \log(x - w^2) + \dots + \log(x - w^{n-1})$$

differentiate the given equation

$$\Rightarrow \frac{(1 + 2x + \dots + (n-1)x^{n-2})}{1 + x + x^2 + \dots + x^{n-1}}$$

$$= \frac{1}{(x - w)} + \frac{1}{(x - w^2)} + \dots + \frac{1}{(x - w^{n-1})}$$

$$\text{put } x = 1$$

$$= \frac{1}{(1 - w)} + \frac{1}{(1 - w^2)} + \dots + \frac{1}{1 - w^{n-1}} = \frac{n-1}{2}$$

5. Ans (A,B)

$$r_1 = 2r_2 = 3r_3$$

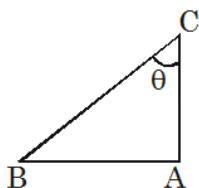
$$\Rightarrow \frac{\Delta}{s-a} = \frac{2\Delta}{s-b} = \frac{3\Delta}{s-c}$$

$$s-a = \lambda, s-b = 2\lambda, s-c = 3\lambda \text{ all these we get}$$

$$s = 6\lambda \text{ so } a = 5\lambda, b = 4\lambda, c = 3\lambda$$

$$(i) \sin^2 A + \sin^2 B + \sin^2 C = 2$$

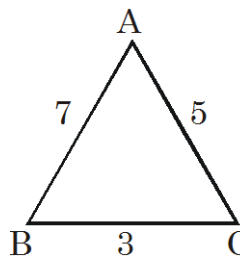
$$(ii) \text{ least angle is } \tan^{-1} \frac{3}{4}$$



$$(iii) \frac{R}{r} = \frac{abc}{4\Delta \cdot \frac{\Delta}{s}}$$

$$= \frac{(abc) \cdot s}{4\Delta^2} = \frac{(5\lambda)(4\lambda)(3\lambda)(6\lambda)}{4 \cdot (6\lambda) \left(\frac{\lambda}{2} \right) (2\lambda)(3\lambda)} = \frac{5}{2}$$

6. Ans (A,B,D)



$$\cos C = \frac{3^2 + 5^2 - 7^2}{2 \cdot 3 \cdot 5} = \frac{9 + 25 - 49}{2 \cdot 15} = -\frac{1}{2}$$

$$\Rightarrow \angle C = \frac{2\pi}{3}$$

$$\Rightarrow$$

$$\tan\left(\frac{A-B}{2}\right) = \left(\frac{a-b}{a+b}\right) \cot \frac{C}{2} = \left(\frac{3-5}{3+5}\right) \cot 60^\circ$$

$$\tan\left(\frac{A-B}{2}\right) = -\frac{1}{4\sqrt{3}}$$

$$\Rightarrow \text{length of median through vertex}$$

$$A = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$$

$$= \frac{1}{2} \sqrt{2 \cdot 5^2 + 2 \cdot 7^2 - 3^2} = \frac{\sqrt{139}}{2}$$

$$\text{length of angle bisector through C}$$

$$= \frac{2ab}{a+b} \cos \frac{C}{2} = \frac{15}{8}$$

MATHEMATICS

SEC-3

1. Ans (B)

P(Red face at any throw) = P(H).P(Red face while throwing A) + P(T).P(Red face while throwing B)

$$P(\text{Red face}) = \frac{1}{2} \times \frac{4}{6} + \frac{1}{2} \times \frac{2}{6} = \frac{1}{2}$$

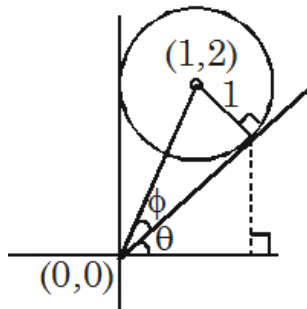
2. Ans (D)

$$\rho = \frac{\left(\frac{1}{2} \cdot \frac{4}{6}\right)^2}{\left(\frac{1}{2} \cdot \frac{4}{6}\right)^2 + \left(\frac{1}{2} \cdot \frac{2}{6}\right)^2 + \left(\frac{1}{2}\right) \left(\frac{4}{6}\right) \left(\frac{1}{2}\right) \left(\frac{2}{6}\right) \cdot 2}$$

$$= \frac{16}{16 + 16 + 4} = \frac{16}{36} = \frac{4}{9}$$

3. Ans (A)

$$\tan(\theta + \phi) = 2 \text{ and } \tan \phi = \frac{1}{2}$$



$$\Rightarrow \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi} = 2$$

$$\tan \theta + \frac{1}{2} = 2 \left(1 - \frac{\tan \theta}{2} \right)$$

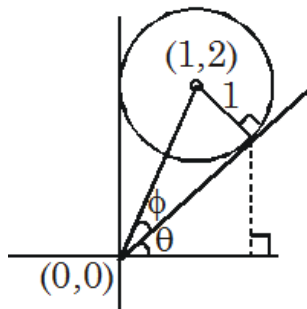
$$\tan \theta = \frac{3}{4} \Rightarrow \theta = \tan^{-1} \frac{3}{4} = (\arg z)_{\min}$$

max distance of the point (3,0) from the circle

$$2\sqrt{2} + 1$$

4. Ans (B)

$$\tan(\theta + \phi) = 2 \text{ and } \tan \phi = \frac{1}{2}$$



$$\Rightarrow \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi} = 2$$

$$\tan \theta + \frac{1}{2} = 2 \left(1 - \frac{\tan \theta}{2} \right)$$

$$\tan \theta = \frac{3}{4} \Rightarrow \theta = \tan^{-1} \frac{3}{4} = (\arg z)_{\min}$$

max distance of the point (3,0) from the circle

$$2\sqrt{2} + 1$$

MATHEMATICS

SEC-4

1. Ans (231)

$$(P) 1 + w + 2w^2 + 3w^3 + \dots + 2012w^{2012}$$

$$(1 + w + 2w^2) + (3w^3 + 4w^4 + 5w^5) + \dots +$$

671 brackets

$$w^2 + 670(w + 2w^2) = 671w^2 - 670$$

$$a = 671, b = -670$$

$$(Q) \lambda = 5 \times 5 \times 2 = 50$$

(R)

$$1 - \left({}^nC_1 \left(\frac{1}{2} \right)^{n-1} \frac{1}{2} + {}^nC_0 \left(\frac{1}{2} \right)^n \right) > .9$$

$$\frac{1}{10} > \frac{(n+1)}{2^n} \Rightarrow 2^n > (n+1) \cdot 10$$

$$n_{\min} = 7$$

2. Ans (312)

$$(P) ||z - 3| - |z + 1|| = 2$$

$$\Rightarrow |2z - 2| = |z - 3 + z - 1|$$

$$\Rightarrow |2z - 2| > ||z - 3| - |z - 1|| > 2$$

$$\Rightarrow |z - 1| > 1$$

(Q)

$$P(n) = (\text{HH} \dots \text{H}) + (\text{TT} \dots \text{T}) = \frac{1}{2^n} + \frac{1}{2^n} = \frac{1}{2^{n-1}}$$

$$\sum_{k=1}^{\infty} \frac{1}{2^{k-1}} = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots = 2$$

$$(R) |z - 2| = \min \{|z|, |z - 4|\}$$

Case -1

$$|z| < |z - 4|, |z - 2| = |z| \Rightarrow x = 1$$

Case -2

$$|z| > |z - 4|, |z - 2| = |z - 4| \Rightarrow x = 3$$

$$\text{sum of Re}(z) = 4$$

3. **Ans (132)**

$$\begin{aligned}
 \text{(P)} \quad 3^{400} &= (9)^{200} = (10 - 1)^{200} \\
 &\Rightarrow {}^{200}C_0 10^{200} - {}^{200}C_1 \cdot 10^{199} + \dots + \\
 &{}^{200}C_{198} \cdot 10^2 - {}^{200}C_{199} \cdot 10 + 1 \\
 &= 100\lambda + 1
 \end{aligned}$$

$$\text{remainder} = 1$$

$$\begin{aligned}
 \text{(Q)} \quad \sum_{r=0}^{n-1} \left(\frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} \right)^3 &= \frac{4}{5} \\
 \Rightarrow \sum_{r=0}^{n-1} \left(\frac{{}^nC_r}{{}^{n+1}C_{r+1}} \right)^3 &= \frac{4}{5}
 \end{aligned}$$

$$\Rightarrow$$

$$\frac{1}{(n+1)^3} \left((n)^3 + (n-1)^3 + \dots + 1^3 \right) = \frac{4}{5}$$

$$\Rightarrow \frac{n^2(n+1)^2}{4(n+1)^3} = \frac{4}{5}$$

$$\Rightarrow 5n^2 + 6n - 16 = 0 \Rightarrow n = 4$$

$$\text{(R)} \quad \left(4^{\frac{1}{3}} + 6^{-\frac{1}{4}} \right)^{20} :$$

$$T_{r+1} = {}^{20}C_r \left(4^{\frac{1}{3}} \right)^{20-r} \left(6^{-\frac{1}{4}} \right)^r$$

$$T_{r+1} = {}^{20}C_r \cdot 4^{\frac{20-r}{3}} \cdot 6^{-\frac{r}{4}}$$

$$\frac{20-r}{3} \in \mathbb{I} \text{ and } \frac{r}{4} \in \mathbb{I}$$

$$r = 8, 20, \text{ No. of rational term} = 2$$

4. **Ans (231)**

$$\text{(P)} \quad f(1) > f(2) > f(3) > f(4) > f(5)$$

$${}^{15}C_5 \text{ (they arrange in only one way)}$$

$$\text{(Q)} \quad f(i) \neq f(j) \text{ for } i \neq j$$

Number of one-one function is

$${}^{15}C_5 \cdot 5!$$

$$\text{(R)} \quad 1 < 2 < 3 < 4 < 5$$

$$\Rightarrow f(1) \leq f(2) \leq f(3) \leq f(4) \leq f(5)$$

$$\Rightarrow {}^4C_0 {}^{15}C_5 + {}^4C_1 {}^{15}C_4 + {}^4C_2 {}^{15}C_3 + {}^4C_3 {}^{15}C_2 +$$

$${}^4C_4 {}^{15}C_1$$

$$\Rightarrow {}^{19}C_5$$